

RF and Microwave Engineering Numerical Companion

EX 716 (BEI) / EX 752 (BEX) · IV-I / IV-II · TU, IOE · Chapter 3, all 12 PYQ S-parameter problems, 2069–2082 BS

3. RF and M/W Network Theory and Analysis: Numerical Problems

12 solved PYQ problems · **≈100 marks** across 11 papers · 2 numeric matrices, 3 derivations, 3 identifications, 3 design problems, 1 modulator

READ THIS FIRST

Chapter 3 numerals are not arithmetic problems. Every one of them is the same four tests of §3.3 applied to a matrix, or the same five-step derivation run backwards to build one. **Nothing here needs a calculator beyond $\sqrt{2}$, a square and a log.** Every number below was computed and asserted in `src/sparam.py`, which re-derives each device matrix from its stated properties rather than quoting it.

- **Always run the four tests in the same order, and always write them as a table.** Examiners give a mark per test; a paragraph hides them.
 - **A matrix that is not unitary is not wrong**, it is lossy. Say so, and say how much: the shortfall in a column is the dissipated fraction.
 - **Watch the normalising factor.** Papers print magic-tee matrices without the $1/\sqrt{2}$. Spotting that it is missing is worth a mark on its own.
 - **Sign, not port number, names the arm.** A column whose entries flip sign is the E-arm (difference); a column whose entries match is the H-arm (sum). Papers swap which port number carries which.
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M. The Four Methods, in Brief

M1: the four tests (Problems 1, 6, 7, 8)

1. **Diagonal:** which $S_{ii} = 0$? Those ports are matched.
2. **Transpose:** is $S_{ij} = S_{ji}$ for every pair? Yes means reciprocal (passive, isotropic); no means ferrite or active.
3. **Column sums:** is $\sum_k |S_{ki}|^2 = 1$ for every column? Yes means lossless; the shortfall is the dissipated fraction of the incident power.
4. **Column dots:** is $\sum_k S_{ki} S_{kj}^* = 0$ for every $i \neq j$? This is the phase half of the unitary condition.

Always write M1 as this table

Test	Computation	Conclusion
Diagonal	every S_{ii}	matched / not
Transpose	S_{ij} vs S_{ji}	reciprocal / not
Column sums	$\sum_k S_{ki} ^2$	lossless, or lossy by the shortfall
Column dots	$\sum_k S_{ki} S_{kj}^*$	phases consistent / not

M2: build a device matrix from its physics (Problems 3, 4, 5)

1. Write the blank $n \times n$ matrix.
2. Impose the **phase facts:** which arms come out in phase ($S_{2k} = +S_{1k}$) and which in antiphase ($S_{2k} = -S_{1k}$).
3. Impose the **matching facts:** $S_{kk} = 0$ for every port the question calls “perfectly matched to the junction”.
4. Impose **reciprocity**, $S_{ij} = S_{ji}$, and count what is left. Both a plane tee and a magic tee leave exactly **4 unknowns**.
5. Apply the **unitary property** $[S][S]^* = [I]$ and read off the equations row by row. Each row-column product gives one equation; solve.

M3: reduce an n -port by terminating ports (Problem 9)

1. A termination on port k forces $a_k = \Gamma_k b_k$, with $\Gamma = -1$ for a **short**, $+1$ for an **open**, 0 for a **matched load**, all measured at the junction reference plane.
2. Write out $b_i = \sum_j S_{ij} a_j$ for every port.
3. Substitute the constraint for each terminated port and eliminate its a and b .
4. What is left is the smaller matrix seen from the remaining ports.

M4: power bookkeeping (Problems 1, 10, 11)

1. Write the drive vector $[a]$ that the connection creates. One transmitter on port 1 alone is $[a] = [a_1, 0, 0, 0]^T$; two equal in-phase transmitters on ports 1 and 2 is $[a] = [a, a, 0, 0]^T$.
2. Multiply: $[b] = [S][a]$.
3. Power leaving port i is $|b_i|^2$, in the same units as $|a|^2$.
4. Check the books: $\sum_i |b_i|^2 \leq \sum_j |a_j|^2$, with equality for a lossless network.

The conversions used throughout

- $RL = -20 \log_{10} |S_{11}|$ dB, $IL = -20 \log_{10} |S_{21}|$ dB.
- $G = +20 \log_{10} |S_{21}|$ dB, $VSWR = \frac{1 + |S_{11}|}{1 - |S_{11}|}$.
- $P_{\text{ref}} = |S_{11}|^2 P_{\text{inc}}$, $P_{\text{trans}} = |S_{21}|^2 P_{\text{inc}}$.
- $P_{\text{diss}} = (1 - \sum_k |S_{k1}|^2) P_{\text{inc}}$.
- $1/\sqrt{2} = 0.7071$, and $(1/\sqrt{2})^2 = 0.5$: half the power.

Problem 1: The four tests on a numeric two-port matrix 3+1+1+1+2

79 Bh · the only fully numeric S-matrix in the archive

QUESTION AS PRINTED 79 Bh [3+1+1+1+2]

Why are S-parameters used in high frequencies? The S-matrix of a certain microwave network is given as $[S] = \begin{bmatrix} 0.4 + j0.5 & j0.6 \\ j0.6 & 0.4 - j0.5 \end{bmatrix}$.

(a) Is the network reciprocal? (b) Is the network lossless? (c) What is the return loss at the input? (d) If the input power to the network is 5 watts, what is the reflected power?

Given

- $S_{11} = 0.4 + j0.5$, $S_{12} = S_{21} = j0.6$, $S_{22} = 0.4 - j0.5$.
- Input power $P_{in} = 5$ W at port 1; port 2 terminated in Z_0 .

Step 0: the magnitudes, computed once and reused

- $|S_{11}| = \sqrt{0.4^2 + 0.5^2} = \sqrt{0.16 + 0.25} = \sqrt{0.41} = 0.6403$.
- $|S_{22}| = \sqrt{0.4^2 + (-0.5)^2} = \sqrt{0.41} = 0.6403$ (the same).
- $|S_{12}| = |S_{21}| = |j0.6| = 0.6$.
- Squares, which is what every power test needs: $|S_{11}|^2 = |S_{22}|^2 = 0.41$, $|S_{12}|^2 = |S_{21}|^2 = 0.36$.

Step 1: run the four tests

Test	Computation	Result	Conclusion
Diagonal	$S_{11} = 0.4 + j0.5 \neq 0$, $S_{22} = 0.4 - j0.5 \neq 0$	both non-zero	Neither port is matched
Transpose	$S_{12} = j0.6$ and $S_{21} = j0.6$	$S_{12} = S_{21}$	(a) Reciprocal ✓
Column 1 sum	$ S_{11} ^2 + S_{21} ^2 = 0.41 + 0.36$	$0.77 \neq 1$	(b) Not lossless
Column 2 sum	$ S_{12} ^2 + S_{22} ^2 = 0.36 + 0.41$	$0.77 \neq 1$	Same shortfall at port 2
Column dot	$S_{11}S_{12}^* + S_{21}S_{22}^* = (0.4 + j0.5)(-j0.6) + (j0.6)(0.4 + j0.5)$ $= (0.30 - j0.24) + (-0.30 + j0.24)$	0	Phases are consistent; only the magnitudes fail

- (a) $S_{12} = S_{21} = j0.6$, so the network is **reciprocal**: it is passive and contains no ferrite and no active device.
- (b) Both columns sum to 0.77, not 1, so the network is **not lossless**. It **dissipates** $1 - 0.77 = 0.23$, i.e. **23%** of whatever is fed into either port.
- Worth adding: the column-dot test *passes*. A network can have the right phase relations and still be lossy, which is why both halves of the unitary condition must be checked.

Step 2: (c) return loss at the input

- Return loss compares incident with reflected power at port 1, and S_{11} is an **amplitude** ratio, so the factor is 20:

$$RL = -20 \log_{10} |S_{11}| = -20 \log_{10}(0.6403)$$

- $\log_{10}(0.6403) = -0.19361$, so $RL = -20 \times (-0.19361)$.
- (c) **$RL = 3.87 \text{ dB}$** .
- Sanity check: a low return loss means a **bad** match, and indeed $VSWR = \frac{1 + 0.6403}{1 - 0.6403} = \frac{1.6403}{0.3597} = 4.56$, far from 1.

Step 3: (d) reflected power for 5 W in

- Power scales as the **square** of the wave amplitude:

$$P_{ref} = |S_{11}|^2 P_{in} = 0.41 \times 5$$

- (d) **$P_{ref} = 2.05 \text{ W}$** .

Step 4: complete the power budget (free marks)

Component	Formula	Value	Share of 5 W
Reflected at port 1	$ S_{11} ^2 P_{in} = 0.41 \times 5$	2.05 W	41%
Delivered to port 2	$ S_{21} ^2 P_{in} = 0.36 \times 5$	1.80 W	36%
Dissipated inside	$(1 - 0.77) \times 5$	1.15 W	23%
Total		5.00 W	100%

- The three rows sum to the input, which is the arithmetic check that the whole answer is consistent.
- Insertion loss, if asked: $IL = -20 \log_{10}(0.6) = \mathbf{4.44 \text{ dB}}$.

Answer to the theory rider (3 marks)

- S-parameters are used at high frequencies because **total V and I are not uniquely defined** when the circuit is comparable to λ , because **open and short terminations cannot be realised** broadband and would **destroy or oscillate** active devices, and because **power** is what microwave instruments actually measure. See §3.1.

Problem 2: Comparing two amplifier S-parameter sets 4

82 Bh · $Z_0 = 50 \Omega$; a reading question, not a design question

QUESTION AS PRINTED 82 Bh [4]

Find the basic differences of the systems having the following sets of S-matrix.

- (a) $S_{11} = 0.65\angle 146^\circ$, $S_{12} = 0.12\angle 46^\circ$, $S_{21} = 2.30\angle 44^\circ$, $S_{22} = 0.17\angle -172^\circ$
 (b) $S_{11} = 0.30\angle -134^\circ$, $S_{12} = 0.03\angle 46^\circ$, $S_{21} = 2.20\angle 44^\circ$, $S_{22} = 0.10\angle -172^\circ$

Given

- Two **two-port** sets, quoted in polar form. Only the **magnitudes** matter for the comparison the question asks for; the angles matter only for Δ and K .

Step 1: what kind of device is each? (the two tests that decide it)

Test	Computation	Result	Conclusion
Transpose	(a) $ S_{12} = 0.12$ vs $ S_{21} = 2.30$; (b) 0.03 vs 2.20	$S_{12} \neq S_{21}$ in both	Both are non-reciprocal
Column 1 sum	(a) $0.65^2 + 2.30^2 = 0.4225 + 5.29$; (b) $0.30^2 + 2.20^2$	5.71 and 4.93 , both $\gg 1$	Both are active: they add power

- $|S_{21}| > 1$ means the device **has gain**, so neither set can describe a passive component. Both are **transistor amplifiers** at the same frequency; the question is which is the better one.

Step 2: convert every magnitude into its engineering figure

Quantity	Formula	System (a)	System (b)	Better
Forward gain	$20 \log_{10} S_{21} $	7.23 dB	6.85 dB	(a), barely
Reverse isolation	$-20 \log_{10} S_{12} $	18.42 dB	30.46 dB	(b), by 12 dB
Input return loss	$-20 \log_{10} S_{11} $	3.74 dB	10.46 dB	(b)
Input VSWR	$\frac{1+ S_{11} }{1- S_{11} }$	4.71	1.86	(b)
Output return loss	$-20 \log_{10} S_{22} $	15.39 dB	20.00 dB	(b)
Output VSWR	$\frac{1+ S_{22} }{1- S_{22} }$	1.41	1.22	(b)

- Sample line, so the arithmetic is visible: $-20 \log_{10}(0.65) = -20 \times (-0.18709) = 3.74$ dB, and $\text{VSWR} = (1 + 0.65)/(1 - 0.65) = 1.65/0.35 = 4.71$.

Step 3: stability, the one place the angles are needed

- $\Delta = S_{11}S_{22} - S_{12}S_{21}$, and $K = \frac{1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2}{2|S_{12}S_{21}|}$.
- System (a):**
 - $S_{11}S_{22} = (0.65)(0.17)\angle(146^\circ - 172^\circ) = 0.1105\angle -26^\circ = 0.0993 - j0.0484$.
 - $S_{12}S_{21} = (0.12)(2.30)\angle(46^\circ + 44^\circ) = 0.276\angle 90^\circ = j0.276$.
 - $\Delta = 0.0993 - j0.0484 - j0.276 = 0.0993 - j0.3244$, so $|\Delta| = 0.3393$.
 - $K = \frac{1 - 0.4225 - 0.0289 + 0.1151}{2 \times 0.276} = \frac{0.6637}{0.552} = 1.202$.
- System (b):**
 - $S_{11}S_{22} = (0.30)(0.10)\angle(-134^\circ - 172^\circ) = 0.03\angle -54^\circ = 0.0176 + j0.0243$.
 - $S_{12}S_{21} = (0.03)(2.20)\angle 90^\circ = j0.066$.
 - $\Delta = 0.0176 - j0.0417$, so $|\Delta| = 0.0453$.
 - $K = \frac{1 - 0.09 - 0.01 + 0.0021}{2 \times 0.066} = \frac{0.9021}{0.132} = 6.834$.
- Both have $K > 1$ and $|\Delta| < 1$, so both are **unconditionally stable**, but (b) is stable by a wide margin and (a) only just.

Answer: the basic differences

- Both are **active, non-reciprocal two-ports**, i.e. amplifiers of nearly the same gain (7.23 dB against 6.85 dB, a difference of 0.4 dB that is within normal device spread).
- The **real difference is everywhere except the gain**:

- **(b) is far better matched at the input:** RL 10.46 dB against 3.74 dB, VSWR 1.86 against 4.71. System (a) reflects 42% of the drive power straight back at the source.
- **(b) is far better isolated:** $|S_{12}| = 0.03$ against 0.12, an extra 12 dB. System (b) is close to **unilateral**, so it can be designed with the simple unilateral equations; system (a) needs the full bilateral treatment.
- **(b) is more stable:** $K = 6.83$ against 1.20, and $|\Delta| = 0.045$ against 0.339. System (a) sits near the edge of unconditional stability, so a mismatched load could push it into oscillation.
- **Verdict:** (b) is the better amplifier at a cost of 0.4 dB of gain: better match, better isolation, much better stability margin.
- **Note for Chapter 6:** the same paper then asks for the maximum unilateral and bilateral gains of set (a). For the record they are $G_{TU,\max} = 9.75$ dB and $G_{\max} = 10.11$ dB; the derivation belongs to the amplifier chapter.

Problem 3: Derive the S-matrix of an E-plane Tee 4 8

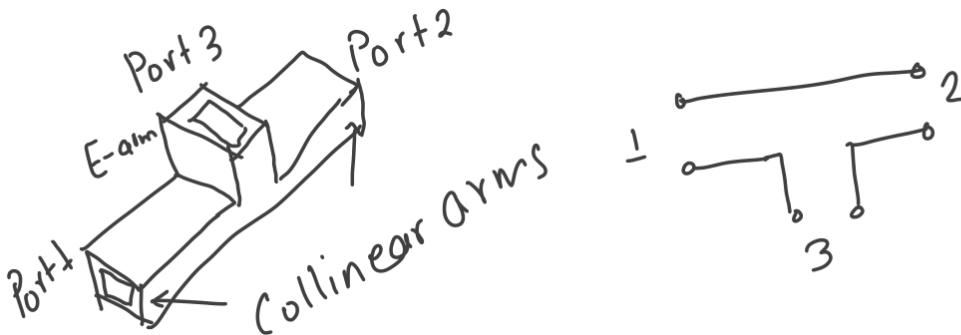
80 Bh, 81 Ba · the model 3-port derivation; learn this and Problems 4 and 5 are the same script

QUESTION AS PRINTED 80 Bh [4+4]

Briefly explain the need of S-parameters and their properties in microwave analysis. Explain three-port network parameters using an example of an E-plane Tee.

Given

- An E-plane (series) tee: ports 1 and 2 are the **collinear arms**, port 3 is the **E-arm** side arm, assumed **perfectly matched to the junction**.
- The junction is a passive, lossless, isotropic waveguide structure.



E-plane tee: 3D view and schematic, ports labelled as used below. Source: RF Pulchowk, S-parameter derivation sheets.

Step 1: write the blank matrix

- A 3-port has $3^2 = 9$ coefficients:

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{bmatrix} \quad (i)$$

Step 2: impose the three physical facts

Physical fact	Equation	Why
An input at the E-arm leaves ports 1 and 2 180° out of phase	$S_{23} = -S_{13}$ (ii)	The E-field of the side arm drives the two collinear arms in opposition
Port 3 is perfectly matched to the junction	$S_{33} = 0$ (iii)	Nothing incident on the E-arm is reflected
The junction is passive and isotropic	$S_{ij} = S_{ji}$ (iv)	Reciprocity: $S_{12} = S_{21}$, $S_{23} = S_{32}$, $S_{31} = S_{13}$

- Substituting (ii), (iii) and (iv) into (i):

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{12} & S_{22} & -S_{13} \\ S_{13} & -S_{13} & 0 \end{bmatrix} \quad (v)$$

- **Four unknowns remain:** $S_{11}, S_{12}, S_{22}, S_{13}$. Four equations are needed, and the unitary property supplies them.

Step 3: apply the unitary (lossless) property

- A lossless network satisfies $[S][S]^*{}^T = [I]$, and since (v) is symmetric this is simply $[S][S]^* = [I]$:

$$\begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{12} & S_{22} & -S_{13} \\ S_{13} & -S_{13} & 0 \end{bmatrix} \begin{bmatrix} S_{11}^* & S_{12}^* & S_{13}^* \\ S_{12}^* & S_{22}^* & -S_{13}^* \\ S_{13}^* & -S_{13}^* & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Step 4: read off one equation per row-column product

Product	Equation	What it gives	
$R_3 \cdot C_3$	$ S_{13} ^2 + S_{13} ^2 = 1 \Rightarrow 2 S_{13} ^2 = 1$	$S_{13} = \frac{1}{\sqrt{2}}$	(vi)
$R_3 \cdot C_1$	$S_{13}S_{11}^* - S_{13}S_{12}^* + 0 = 0 \Rightarrow S_{11}^* = S_{12}^*$	$S_{11} = S_{12}$	(vii)
$R_1 \cdot C_1$	$ S_{11} ^2 + S_{12} ^2 + S_{13} ^2 = 1$, and with (vi) and (vii) $2 S_{11} ^2 = 1 - \frac{1}{2}$	$ S_{11} ^2 = \frac{1}{4}$, so $S_{11} = S_{12} = \frac{1}{2}$	
$R_2 \cdot C_2$	$ S_{12} ^2 + S_{22} ^2 + S_{13} ^2 = 1$, i.e. $\frac{1}{4} + S_{22} ^2 + \frac{1}{2} = 1$	$ S_{22} ^2 = \frac{1}{4}$, so $S_{22} = \frac{1}{2}$	

Answer

- Substituting all four values back into (v):

$$[S]_{\text{E-tee}} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \end{bmatrix}$$

Reading the answer back (this is the “explain” half of the marks)

- **Power split:** $|S_{13}|^2 = |S_{23}|^2 = \frac{1}{2}$, so an input at the E-arm divides **equally** between the collinear arms.
- **Phase:** $S_{23} = -S_{13}$, so those two outputs are **antiphase**; this is what makes the E-tee a **difference** junction.
- **Matching:** $S_{33} = 0$ but $S_{11} = S_{22} = \frac{1}{2}$, so the collinear arms are **not matched**: a quarter of the power incident on port 1 comes straight back. That is the 3-port theorem of §3.5 appearing as arithmetic.
- **Isolation:** $S_{12} = \frac{1}{2} \neq 0$, so the collinear arms are **not isolated** from each other either. Adding the H-arm to fix both defects is exactly what turns a tee into a magic tee (Problem 5).
- **Check:** every column sums to $\frac{1}{4} + \frac{1}{4} + \frac{1}{2} = 1$ ✓ lossless.

Problem 4: Derive the S-matrix of an H-plane Tee 8

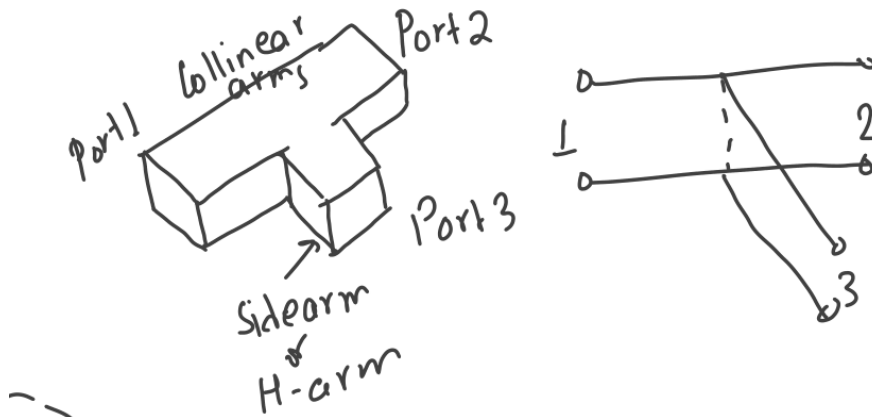
81 Ba · the same script as Problem 3 with one sign changed

QUESTION AS PRINTED 81 Ba [8]

Prepare S-matrices of a perfectly working E-plane Tee, H-plane Tee and a Duplexer, and explore their uses.

Given

- An H-plane (shunt) tee: ports 1 and 2 collinear, port 3 the **H-arm** side arm, perfectly matched to the junction.



H-plane tee: the side arm lies in the plane of the magnetic field. Source: RF Pulchowk, S-parameter derivation sheets.

Step 1: the one fact that differs from Problem 3

- In an H-plane tee an input at the side arm reaches the two collinear arms **in phase**:

$$S_{13} = S_{23} \quad (\text{ii}')$$

- Everything else is unchanged: $S_{33} = 0$ (iii) and $S_{ij} = S_{ji}$ (iv).
- So the working matrix is

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{12} & S_{22} & S_{13} \\ S_{13} & S_{13} & 0 \end{bmatrix} \quad (\text{v}')$$

with the same four unknowns.

Step 2: unitary property, row by row

Product	Equation	What it gives
$R_3 \cdot C_3$	$ S_{13} ^2 + S_{13} ^2 = 1$	$S_{13} = \frac{1}{\sqrt{2}}$
$R_3 \cdot C_1$	$S_{13}S_{11}^* + S_{13}S_{12}^* = 0 \Rightarrow S_{11}^* = -S_{12}^*$	$S_{11} = -S_{12}$
$R_1 \cdot C_1$	$ S_{11} ^2 + S_{12} ^2 + \frac{1}{2} = 1 \Rightarrow 2 S_{11} ^2 = \frac{1}{2}$	$S_{11} = \frac{1}{2}$, so $S_{12} = -\frac{1}{2}$
$R_2 \cdot C_2$	$\frac{1}{4} + S_{22} ^2 + \frac{1}{2} = 1$	$S_{22} = \frac{1}{2}$

- **The sign moved.** In the E-tee the minus sign sat in the side-arm column; here it sits in S_{12} instead. That single difference is the whole distinction between the two tees.

Answer, and the duplexer the same question asks for

- H-plane tee:

$$[S]_{\text{H-tee}} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{\sqrt{2}} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix}$$

- **Duplexer**, taken as an ideal 3-port ferrite circulator with port 1 the transmitter, port 2 the antenna and port 3 the receiver:

$$[S]_{\text{duplexer}} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Matched at every port ($S_{ii} = 0$), lossless (every column sums to 1) and **deliberately non-reciprocal** ($S_{21} = 1$ but $S_{12} = 0$), which is the only way a 3-port can be both matched and lossless (§3.5).

Uses, which the question asks for explicitly

Device	Behaviour	Uses
E-plane tee	Equal split, outputs in antiphase; difference junction	Balanced mixers, phase comparators, the difference arm of a magic tee, feeding balanced antennas
H-plane tee	Equal split, outputs in phase; sum junction	In-phase power dividers and combiners, feeding two antennas from one source, the sum arm of a magic tee
Duplexer	One antenna shared by transmitter and receiver, with the receiver isolated from the transmitter	Radar, and any single-antenna transceiver; protects the sensitive receiver front end from the transmit pulse

Problem 5: Derive the S-matrix of a Magic Tee 3+3 8 10

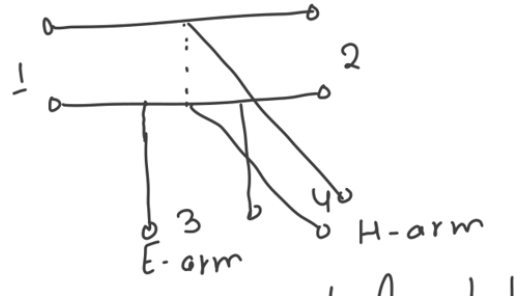
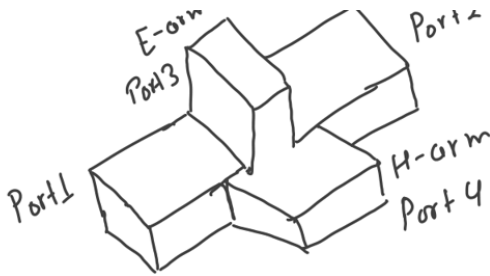
72 Ash, 74 Bh, 75 Bh, 78 Ch, 81 Bh · the single highest-value derivation in the chapter

QUESTION AS PRINTED The device behind five identification questions

Describe the operational principles of a magic tee based on its S-parameters, and derive its scattering matrix.

Given

- A hybrid (magic) tee: ports 1 and 2 are the **collinear arms**, port 3 is the **E-arm** and port 4 is the **H-arm**. Both side arms are perfectly matched to the junction.



Magic tee: an E-plane tee and an H-plane tee sharing the same junction. Source: RF Pulchowk, S-parameter derivation sheets.

Step 1: write the blank 4×4

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{21} & S_{22} & S_{23} & S_{24} \\ S_{31} & S_{32} & S_{33} & S_{34} \\ S_{41} & S_{42} & S_{43} & S_{44} \end{bmatrix} \quad (\text{i})$$

Step 2: impose the five physical facts

Physical fact	Equation	Why
It contains an E-plane tee, so an input at the E-arm (port 3) leaves ports 1 and 2 antiphase	$S_{23} = -S_{13}$	(ii) Same argument as Problem 3
It contains an H-plane tee, so an input at the H-arm (port 4) leaves ports 1 and 2 in phase	$S_{24} = +S_{14}$	(iii) Same argument as Problem 4
An input at the E-arm produces no output at the H-arm, and the reverse	$S_{34} = S_{43} = 0$	(iv) The two side arms are geometrically orthogonal; the E-arm excites an antisymmetric field, the H-arm a symmetric one
Both side arms are perfectly matched to the junction	$S_{33} = S_{44} = 0$	(v) Stated in the question
No output at port 2 for an input at port 1, and the reverse	$S_{12} = S_{21} = 0$	(vi) The collinear arms are decoupled, which is the “magic”

- With reciprocity $S_{ij} = S_{ji}$ (vii), equations (ii) to (vii) reduce (i) to

$$[S] = \begin{bmatrix} S_{11} & 0 & S_{13} & S_{14} \\ 0 & S_{22} & -S_{13} & S_{14} \\ S_{13} & -S_{13} & 0 & 0 \\ S_{14} & S_{14} & 0 & 0 \end{bmatrix} \quad (\text{viii})$$

- Four unknowns:** $S_{11}, S_{22}, S_{13}, S_{14}$.

Step 3: unitary property, row by row

Product	Equation	What it gives
$R_3 \cdot C_3$	$ S_{13} ^2 + -S_{13} ^2 + 0 + 0 = 1$, i.e. $2 S_{13} ^2 = 1$	$S_{13} = \frac{1}{\sqrt{2}}$
$R_4 \cdot C_4$	$ S_{14} ^2 + S_{14} ^2 + 0 + 0 = 1$, i.e. $2 S_{14} ^2 = 1$	$S_{14} = \frac{1}{\sqrt{2}}$
$R_1 \cdot C_1$	$ S_{11} ^2 + 0 + S_{13} ^2 + S_{14} ^2 = 1$, i.e. $ S_{11} ^2 + \frac{1}{2} + \frac{1}{2} = 1$	$ S_{11} ^2 = 0$, so $S_{11} = 0$
$R_2 \cdot C_2$	$0 + S_{22} ^2 + S_{13} ^2 + S_{14} ^2 = 1$	$S_{22} = 0$

- **This is the striking result:** matching the two side arms and decoupling the collinear pair **forces** the collinear arms to be matched as well. Nothing was assumed about S_{11} or S_{22} ; the unitary condition produced them.

Answer

$$[S]_{\text{magic tee}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

Operational principles, read straight off the matrix

Drive	Outputs	Consequence
E-arm (port 3)	$b_1 = \frac{1}{\sqrt{2}}a_3, b_2 = -\frac{1}{\sqrt{2}}a_3,$ $b_4 = 0$	Equal split, antiphase, nothing into the H-arm
H-arm (port 4)	$b_1 = b_2 = \frac{1}{\sqrt{2}}a_4, b_3 = 0$	Equal split, in phase, nothing into the E-arm
Port 1	$b_3 = \frac{1}{\sqrt{2}}a_1, b_4 = \frac{1}{\sqrt{2}}a_1,$ $b_2 = 0$	Splits into both side arms, nothing reaches port 2
Ports 1 and 2 together	$b_4 \propto (a_1 + a_2),$ $b_3 \propto (a_1 - a_2)$	H-arm is the sum arm, E-arm the difference arm

- **Why “magic”:** all four ports are matched *and* two pairs (1-2 and 3-4) are completely isolated, which a 3-port can never achieve.
- **Uses:** power combining and splitting (§3.8), balanced mixers, duplexers, impedance bridges, monopulse radar comparators.
- **In practice** the ideal matrix needs matching posts or irises at the junction; without them S_{11} and S_{22} are not zero.

Problem 6: Identify the device from a numeric 4-port matrix 10

78 Ch · the printed matrix is missing its normalising factor

QUESTION AS PRINTED 78 Ch [10]

Identify and explain the properties of a microwave passive device having the following S-matrix, with necessary diagram. $[S] =$

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

Given

- A 4×4 matrix, so the device has **four ports**.
- **The paper prints the first row as “0.0 1 1”;** the stray point is a scanner artefact over a zero, and the matrix is the standard form.

Step 1: run the four tests

Test	Computation	Result	Conclusion
Diagonal	$S_{11} = S_{22} = S_{33} = S_{44} = 0$	all zero	Matched at every port
Transpose	$S_{13} = S_{31} = 1, S_{23} = S_{32} = -1,$ $S_{14} = S_{41} = 1, S_{24} = S_{42} = 1$	$[S] = [S]^T$	Reciprocal , so passive and non-ferrite
Column sums	$ 0 ^2 + 0 ^2 + 1 ^2 + 1 ^2 = 2$ for every column	$2 \neq 1$	Not unitary as printed : the $1/\sqrt{2}$ factor has been omitted
Column dots	$C_1 \cdot C_2^* = (0)(0) + (0)(0) + (1)(-1) + (1)(1) = 0;$ $C_3 \cdot C_4^* = (1)(1) + (-1)(1) + 0 + 0 = 0$	all zero	Phases are consistent; only the scale is wrong

- **Judging the condition**: divide the printed matrix by $\sqrt{2}$. Then every column sums to $\frac{1}{2} + \frac{1}{2} = 1$ and the device is **lossless**. **Say this explicitly**: it is the “judge the condition” mark.

Step 2: map the zeros and read the signs

- **Zero blocks**: $S_{12} = S_{21} = 0$, so **ports 1 and 2 are isolated**; $S_{34} = S_{43} = 0$, so **ports 3 and 4 are isolated**. Two isolated pairs on a matched, lossless, reciprocal 4-port.
- **Column 3** reads $(1, -1)$ down the collinear ports: the signs **flip**, so port 3 is the **difference (E) arm**.
- **Column 4** reads $(1, 1)$: the signs **match**, so port 4 is the **sum (H) arm**.
- **Equal magnitudes** everywhere off the zero blocks, so every split is 50/50.

Answer

- The device is a magic tee (hybrid tee), with ports 1 and 2 the collinear arms, port 3 the E-arm and port 4 the H-arm. Normalised:

$$[S] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

- Diagram: reproduce the 3D sketch of Problem 5 with all four arms labelled.

Properties, one line each

- **Matched** at all four ports, so no port reflects: $S_{ii} = 0$, return loss infinite in the ideal case.
- **Reciprocal**, $[S] = [S]^T$: passive waveguide, no ferrite, no gain.
- **Lossless** once normalised: all incident power is redistributed, none dissipated.
- **Two isolated pairs**: 1-2 and 3-4. Two sources on ports 1 and 2 cannot see each other, and the E-arm and H-arm are decoupled.
- **Equal power division**, $|S_{13}|^2 = \frac{1}{2}$: every drive splits in half, i.e. a 3 dB hybrid.
- **Phase**: E-arm drive gives antiphase outputs, H-arm drive gives in-phase outputs.
- **Uses**: power combiner and splitter, duplexer, balanced mixer, monopulse comparator, impedance bridge.

Problem 7: Identify and complete a symbolic 4-port matrix 5 8

74 Bh, 81 Bh · the same device as Problem 6, printed before the unitary step

QUESTION AS PRINTED 81 Bh [5] and 74 Bh [8]

Identify and explain, along with a neat diagram, the properties of the passive microwave device described by the following S-matrix:

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{12} & S_{22} & S_{13} & -S_{14} \\ S_{13} & S_{13} & 0 & 0 \\ S_{14} & -S_{14} & 0 & 0 \end{bmatrix}$$

Step 1: read the structure before doing any algebra

Feature	Where	What it means
$S_{33} = S_{44} = 0$	bottom-right block	Ports 3 and 4 are matched to the junction
$S_{34} = S_{43} = 0$	bottom-right block	Ports 3 and 4 are isolated from each other
Column 3 is (S_{13}, S_{13})	signs match	Port 3 drives the collinear arms in phase : the H-arm
Column 4 is $(S_{14}, -S_{14})$	signs flip	Port 4 drives them in antiphase : the E-arm
$[S] = [S]^T$	everywhere	Reciprocal, hence passive and isotropic

- Four ports, two matched side arms, one sum arm and one difference arm: **this is a magic tee**. Note that the port labels are the reverse of Problem 6: here **port 3 is the H-arm and port 4 the E-arm**. **Read the signs, not the numbers**.

Step 2: finish the job the printed matrix leaves undone

- The matrix still carries S_{11} , S_{12} , S_{22} , S_{13} , S_{14} as unknowns. Apply the unitary property exactly as in Problem 5:

Product	Equation	Result
$R_3 \cdot C_3$	$ S_{13} ^2 + S_{13} ^2 = 1$	$S_{13} = \frac{1}{\sqrt{2}}$
$R_4 \cdot C_4$	$ S_{14} ^2 + -S_{14} ^2 = 1$	$S_{14} = \frac{1}{\sqrt{2}}$
$R_3 \cdot C_4$	$S_{13}S_{14}^* - S_{13}S_{14}^* = 0$	satisfied identically, no new information
$R_1 \cdot C_2$	$S_{11}S_{12}^* + S_{12}S_{22}^* + S_{13} ^2 - S_{14} ^2 = 0$ and the last two cancel	$S_{11}S_{12}^* + S_{12}S_{22}^* = 0$
$R_1 \cdot C_1$	$ S_{11} ^2 + S_{12} ^2 + \frac{1}{2} + \frac{1}{2} = 1$	$ S_{11} ^2 + S_{12} ^2 = 0$, so $S_{11} = S_{12} = 0$
$R_2 \cdot C_2$	$ S_{12} ^2 + S_{22} ^2 + \frac{1}{2} + \frac{1}{2} = 1$	$S_{22} = 0$

Answer

- Magic tee, with the completed matrix

$$[S] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix}$$

which is Problem 6's matrix with columns 3 and 4 interchanged, as the port labelling demands.

- The strongest single line in this answer:** nothing was assumed about S_{11} , S_{12} or S_{22} , yet all three came out zero. Matching the two side arms of a 4-port **forces** the collinear arms to be matched and isolated. That is precisely what a 3-port cannot do (§3.5).
- Properties and uses: as listed in Problem 6.

Problem 8: Judge the condition of a given S-matrix 6

75 Bh · the isolation is already printed in, so only the scale is left to judge

QUESTION AS PRINTED 75 Bh [6]

Which passive microwave device is explained by this S-matrix? Judge the condition and explain its characteristics. $[S] = \begin{bmatrix} S_{11} & 0 & S_{13} & S_{14} \\ 0 & S_{22} & -S_{13} & S_{14} \\ S_{13} & -S_{13} & 0 & 0 \\ S_{14} & S_{14} & 0 & 0 \end{bmatrix}$

Step 1: identify

- Four ports; $S_{33} = S_{44} = 0$ and $S_{34} = S_{43} = 0$, so ports 3 and 4 are matched and mutually isolated.
- $S_{12} = S_{21} = 0$ is **already written in**, so ports 1 and 2 are isolated too: the collinear pair.
- Column 3 is $(S_{13}, -S_{13})$, signs flipping → **E-arm**. Column 4 is (S_{14}, S_{14}) , signs matching → **H-arm**.
- $[S] = [S]^T$ → reciprocal, so passive.
- Magic tee, with the port labelling of Problem 6.

Step 2: judge the condition (what the question actually rewards)

Test	Computation	Result	Conclusion
Diagonal	S_{11}, S_{22} left as unknowns; $S_{33} = S_{44} = 0$	side arms matched	Collinear match not yet stated
Transpose	every $S_{ij} = S_{ji}$	symmetric	Reciprocal
Column 3 sum	$ S_{13} ^2 + -S_{13} ^2 = 1$	$S_{13} = \frac{1}{\sqrt{2}}$	Equal, antiphase split from the E-arm
Column 4 sum	$ S_{14} ^2 + S_{14} ^2 = 1$	$S_{14} = \frac{1}{\sqrt{2}}$	Equal, in-phase split from the H-arm
Column 1 sum	$ S_{11} ^2 + 0 + \frac{1}{2} + \frac{1}{2} = 1$	$S_{11} = 0$	Port 1 is matched after all
Column 2 sum	$0 + S_{22} ^2 + \frac{1}{2} + \frac{1}{2} = 1$	$S_{22} = 0$	Port 2 likewise

- **The condition:** the matrix describes a **lossless, reciprocal, matched-at-all-four-ports** device, and this is **consistent** only because it has four ports. The same set of demands on a 3-port is impossible (§3.5).
- Completed:

$$[S] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

Characteristics

- **3 dB equal division:** $|S_{13}|^2 = |S_{14}|^2 = \frac{1}{2}$, so every input splits half and half.
- **Double isolation:** 1-2 and 3-4. Two transmitters on the collinear arms cannot load each other; the E-arm and H-arm cannot leak into each other.
- **Phase discrimination:** sum at the H-arm, difference at the E-arm, which is what makes it usable as a comparator.
- **Bidirectional:** being reciprocal, it combines as readily as it splits.
- **Limitation:** the ideal matrix assumes matching irises at the junction and holds over a limited bandwidth.

Problem 9: Magic tee with both E- and H-arms short-circuited 4+4

82 Bh · verified/added no source note carries this; derived here and asserted in `sparam.py`

QUESTION AS PRINTED 82 Bh [4+4]

Design a model of a three-port network and define its characteristic parameters. Prepare the S-matrix of a rectangular magic tee having both E- and H-arms shorted, and explain how it behaves.

Given

- The ideal magic tee of Problem 5, with ports 3 (E-arm) and 4 (H-arm) **short-circuited at the junction reference plane**.
- A short reflects with $\Gamma = -1$, so $a_3 = -b_3$ and $a_4 = -b_4$.

Step 1: write out the four scattering equations

- From $[b] = [S][a]$ with the magic-tee matrix:

$$\begin{aligned} b_1 &= \frac{1}{\sqrt{2}}(a_3 + a_4), & b_2 &= \frac{1}{\sqrt{2}}(-a_3 + a_4) \\ b_3 &= \frac{1}{\sqrt{2}}(a_1 - a_2), & b_4 &= \frac{1}{\sqrt{2}}(a_1 + a_2) \end{aligned}$$

Step 2: substitute the short-circuit conditions

- $a_3 = -b_3 = -\frac{1}{\sqrt{2}}(a_1 - a_2)$.
- $a_4 = -b_4 = -\frac{1}{\sqrt{2}}(a_1 + a_2)$.

Step 3: eliminate and collect

- Into the expression for b_1 :

$$b_1 = \frac{1}{\sqrt{2}} \left[-\frac{1}{\sqrt{2}}(a_1 - a_2) - \frac{1}{\sqrt{2}}(a_1 + a_2) \right] = -\frac{1}{2} [(a_1 - a_2) + (a_1 + a_2)] = -\frac{1}{2}(2a_1) = -a_1$$

- Into the expression for b_2 :

$$b_2 = \frac{1}{\sqrt{2}} \left[+\frac{1}{\sqrt{2}}(a_1 - a_2) - \frac{1}{\sqrt{2}}(a_1 + a_2) \right] = \frac{1}{2} [(a_1 - a_2) - (a_1 + a_2)] = \frac{1}{2} (-2a_2) = -a_2$$

- The a_2 terms cancel in b_1 and the a_1 terms cancel in b_2 : the two reflections arriving from the shorted arms add in the driven arm and subtract in the other one.

Answer

- Seen from the two collinear arms, the shorted magic tee is the two-port

$$[S]_{\text{shorted}} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

How it behaves

- **Total reflection:** $|S_{11}| = |S_{22}| = 1$, so every watt fed into port 1 comes back out of port 1. Return loss = 0 dB, VSWR = ∞ .
- **Phase reversal:** the -1 is a 180° shift, exactly what an ideal short does. The tee behaves as a **pair of independent short circuits**, one at each collinear arm.
- **Complete isolation survives:** $S_{12} = S_{21} = 0$, so the two arms still cannot see each other even though both are totally reflecting.
- **Still lossless:** each column sums to $|-1|^2 = 1$; nothing is dissipated, everything is returned.
- **Still reciprocal:** $[S] = [S]^T$.
- **Use:** a reflective phase shifter or a calibration short. Moving the shorts a distance ℓ out from the junction changes Γ from -1 to $-e^{-2j\beta\ell}$, and the same algebra then gives $S_{11} = S_{22} = -e^{-2j\beta\ell}$ with S_{12} still zero, so a tunable phase with a constant unity magnitude.

Problem 10: Two radar transmitters, twice the power into one antenna 6 8

73 Bh, 74 Ma · a design question answered entirely by one matrix multiplication

QUESTION AS PRINTED 73 Bh [8] and 74 Ma [6]

Suppose there are two identical radar transmitters and a few passive devices in equipment stock. A particular application requires twice more input power to an antenna than either transmitter can deliver. As an RF engineer, give your appropriate solution with necessary figures, mathematics and sufficient explanation.

Given

- Two identical transmitters, each able to deliver P watts.
- One antenna, which must receive $2P$ watts.

Step 1: choose the device, and say why

- **Choice:** a **magic tee** used as a power **combiner** (a 3 dB hybrid coupler works identically).
- **Why not a simple T-junction or Y-junction:**
 - A plain junction gives $S_{12} \neq 0$, so each transmitter drives power into the other one: mutual loading, pulling of the oscillators, and possible damage.
 - Its collinear ports are not matched ($S_{11} = S_{22} = \frac{1}{2}$ for a tee, Problem 3), so both transmitters see a high VSWR.
- The magic tee cures both: $S_{12} = 0$ (perfect isolation) and $S_{ii} = 0$ (matched at every port).

Step 2: the connection

Port	Connected to	Reason
1 (collinear)	Transmitter A	Isolated from port 2
2 (collinear)	Transmitter B	Isolated from port 1
3 (E-arm)	Matched dummy load	Takes the difference , which is zero when the drives are balanced
4 (H-arm)	Antenna	Takes the sum of the two drives

- The two feed lines must be cut to the **same electrical length**, so that the transmitters arrive in phase.

Step 3: the mathematics

- Both transmitters equal and in phase, nothing incident on the side arms:

$$[a] = [a \quad a \quad 0 \quad 0]^T, \quad |a|^2 = P$$

- Multiply $[b] = [S][a]$ with the magic-tee matrix:

$$b_3 = \frac{1}{\sqrt{2}}(a_1 - a_2) = \frac{1}{\sqrt{2}}(a - a) = 0$$

$$b_4 = \frac{1}{\sqrt{2}}(a_1 + a_2) = \frac{1}{\sqrt{2}}(2a) = \sqrt{2}a$$

$$b_1 = \frac{1}{\sqrt{2}}(a_3 + a_4) = 0, \quad b_2 = \frac{1}{\sqrt{2}}(-a_3 + a_4) = 0$$

Step 4: read the powers

Port	Wave	Power	Meaning
4, antenna	$b_4 = \sqrt{2}a$	$ b_4 ^2 = 2 a ^2 = 2P$	The requirement, met exactly
3, dummy load	$b_3 = 0$	0	Nothing wasted when the drives are balanced
1 and 2, transmitters	$b_1 = b_2 = 0$	0	No reflection back into either transmitter
Total out		2P	Equals the total in, as a lossless network requires

Answer

- Feed the two transmitters into the collinear arms of a magic tee, antenna on the H-arm, matched load on the E-arm. The antenna then receives $2P$ watts, twice what either transmitter alone can deliver.

Explanation, which carries the remaining marks

- Why the sum appears at the H-arm:** an H-plane junction excites the two collinear arms in phase, so by reciprocity two in-phase drives add there.
- Why the dummy load matters:** if the transmitters are unequal or out of phase, $b_3 = \frac{1}{\sqrt{2}}(a_1 - a_2) \neq 0$ and the imbalance is **absorbed** by the load instead of being reflected into the transmitters. The load must be rated for the worst-case imbalance, i.e. the full power of one transmitter.
- Phase error costs power:** with a phase error ϕ between the drives, the antenna receives $2P \cos^2(\phi/2)$ and the load takes the rest. A 20° error already loses 0.13 dB.
- Alternatives:** a 3 dB branch-line or rat-race hybrid does the same job in microstrip; a Wilkinson combiner does it with an isolation resistor when the device must be planar.

Problem 11: ASR duplexer delivering half the transmitter power 8 10

73 Ma, 76 Bh · the same magic tee, driven from the other end

QUESTION AS PRINTED 76 Bh [10] and 73 Ma [8]

A radar installation engineer is given the responsibility of installing an Airport Surveillance Radar that requires half the input power to the antenna than the transmitter can deliver, using a duplexer. Prepare its S-matrix. (73 Ma: choose a suitable passive microwave device to split power in half and explain its properties.)

Given

- A transmitter delivering P watts.
- The antenna must receive $P/2$ watts, and the receiver must share the same antenna while staying isolated from the transmitter.

Step 1: choose the device

- Choice:** a **magic tee used as a duplexer**. It is the only passive device that splits power exactly in half *and* provides the transmitter-to-receiver isolation a duplexer must have.
- A resistive splitter would also halve the power but would burn the other half and give no isolation.

Step 2: the connection

Port	Connected to	Reason
4 (H-arm)	Transmitter	Drives both collinear arms equally and in phase
1 (collinear)	Antenna	Receives half the transmitter power
2 (collinear)	Matched dummy load	Absorbs the other half, and balances the bridge
3 (E-arm)	Receiver	Isolated from the H-arm, so the transmit pulse never reaches it

Step 3: transmit path

- Drive vector $[a] = [0 \ 0 \ 0 \ a]^T$ with $|a|^2 = P$.
- From $[b] = [S][a]$:

$$b_1 = \frac{1}{\sqrt{2}}a_4 = \frac{1}{\sqrt{2}}a, \quad b_2 = \frac{1}{\sqrt{2}}a_4 = \frac{1}{\sqrt{2}}a, \quad b_3 = 0$$

- Powers:

$$|b_1|^2 = \frac{1}{2}|a|^2 = \frac{P}{2}, \quad |b_2|^2 = \frac{P}{2}, \quad |b_3|^2 = 0$$

- **The antenna receives exactly $P/2$** , which is the requirement, and the receiver on the E-arm receives **nothing** from the transmitter, which is the point of a duplexer.

Step 4: receive path

- The echo arrives at port 1 alone: $[a] = [a_1 \ 0 \ 0 \ 0]^T$ with $|a_1|^2 = P_{echo}$.
- $b_3 = \frac{1}{\sqrt{2}}a_1$ and $b_4 = \frac{1}{\sqrt{2}}a_1$, so

$$|b_3|^2 = \frac{P_{echo}}{2} \text{ to the receiver,} \quad |b_4|^2 = \frac{P_{echo}}{2} \text{ back into the transmitter}$$

- **Nothing reaches port 2**, since $S_{21} = 0$.

Answer

- The duplexer is the magic tee, whose S-matrix is

$$[S] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

with port 4 the transmitter, port 1 the antenna, port 2 the matched load and port 3 the receiver.

Properties and the honest limitation

- **Equal split:** $|S_{14}|^2 = |S_{24}|^2 = \frac{1}{2}$, a 3 dB hybrid.
- **Isolation:** transmitter (port 4) to receiver (port 3) is $S_{34} = 0$, so the receiver front end is protected from the full transmit pulse.
- **Matched at every port**, so neither the transmitter nor the antenna sees a standing wave.
- **The 3 dB penalty is real and should be stated:** half the transmit power goes into the dummy load, and half the echo goes back to the transmitter. The magic-tee duplexer therefore costs 3 dB in each direction.
- **Better in practice:** a **ferrite circulator** duplexer, $[S] = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, routes the *whole* transmit power to the antenna and the *whole* echo to the receiver, with no 3 dB loss. Quote it as the improvement whenever the question does not explicitly demand the half-power split, as **76 Bh** does.

Problem 12: Evaluate the S-matrix of an amplitude modulator 7

78 Ch · verified/added no source note carries this; modelled and evaluated here

QUESTION AS PRINTED 78 Ch [3+7]

Elaborate the importance of S-parameters for microwave frequencies. Evaluate the S-matrix of an Amplitude Modulator.

Step 1: model the modulator as a two-port

- A microwave amplitude modulator sits **in** the RF path: the carrier enters port 1 and the modulated carrier leaves port 2. The modulating signal is a **control input**, applied as bias, not as a third RF port.

- Physically it is a **PIN-diode variable attenuator**: the diode's RF resistance is set by its DC bias, so the modulating waveform steers the transmission through the device.
- Both RF ports are designed to stay **matched** across the modulation swing, otherwise the modulation would pull the source and produce incidental phase modulation.

Step 2: impose the three properties

Property	Equation	Why
Matched at both ports	$S_{11} = S_{22} = 0$	The attenuator is built as a matched pad at every bias point
Reciprocal	$S_{12} = S_{21} = \alpha$	Passive, no ferrite and no gain: a diode resistance is bilateral
Lossy, and time varying	$0 \leq \alpha(t) \leq 1$	The power removed from the carrier is dissipated in the diode

- Hence

$$[S] = \begin{bmatrix} 0 & \alpha(t) \\ \alpha(t) & 0 \end{bmatrix}, \quad \alpha(t) = \alpha_0 (1 + m \cos \omega_m t)$$

with α_0 the carrier-level transmission, m the modulation index and ω_m the modulating frequency.

Step 3: check the condition

- Column sum: $|S_{11}|^2 + |S_{21}|^2 = \alpha^2$, which equals 1 only when $\alpha = 1$.
- So the modulator is **never lossless** except at full transmission: the dissipated fraction is $1 - \alpha^2(t)$, and this is unavoidable. An amplitude modulator works by **throwing power away** in step with the modulating signal.
- It is reciprocal and matched, so the only non-ideal property is the loss.

Step 4: evaluate with numbers

- Take $\alpha_0 = 0.5$ and $m = 0.6$, a typical shallow modulation.

Instant	$\alpha(t)$	$[S]$	IL and P_{out} for 1 W in
Peak, $\cos \omega_m t = +1$	$0.5(1 + 0.6) = 0.80$	$\begin{bmatrix} 0 & 0.8 \\ 0.8 & 0 \end{bmatrix}$	$IL = -20 \log 0.8 = 1.94 \text{ dB}; P_{out} = 0.64 \text{ W}$
Carrier, $\cos \omega_m t = 0$	0.5	$\begin{bmatrix} 0 & 0.5 \\ 0.5 & 0 \end{bmatrix}$	$IL = 6.02 \text{ dB}; P_{out} = 0.25 \text{ W}$
Trough, $\cos \omega_m t = -1$	$0.5(1 - 0.6) = 0.20$	$\begin{bmatrix} 0 & 0.2 \\ 0.2 & 0 \end{bmatrix}$	$IL = 13.98 \text{ dB}; P_{out} = 0.04 \text{ W}$

- **Recovering the modulation index from the matrix**, which is the check the examiner is looking for:

$$m = \frac{|S_{21}|_{\max} - |S_{21}|_{\min}}{|S_{21}|_{\max} + |S_{21}|_{\min}} = \frac{0.80 - 0.20}{0.80 + 0.20} = \frac{0.60}{1.00} = 0.6 \checkmark$$

- **Modulation depth in dB**: the transmission swings over $13.98 - 1.94 = 12.04 \text{ dB}$.

Notes worth adding

- **Balanced realisation**: build the modulator on a magic tee, carrier into the H-arm, matched PIN diodes in the two collinear arms and the modulating bias on the E-arm. The two diodes reflect equally, the reflections recombine into the E-arm and the output amplitude follows the bias. The device is then a magic tee with two variable terminations, and the algebra is Problem 9 with Γ set by the bias instead of fixed at -1 .
- S_{11} **is the figure of merit**: a good modulator keeps S_{11} small at every bias point. If it does not, the source sees a varying load and the output picks up incidental phase modulation.
- **Contrast with an ideal switch**: α toggling between 1 and 0 makes the same matrix an on-off (ASK) modulator, with $m = 1$.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
RF	radio frequency	MW, M/W	microwave
EM, EMR	electromagnetic (radiation)	TEM	transverse electromagnetic mode
TE	transverse electric mode	TM	transverse magnetic mode
WG	waveguide	TL	transmission line
VSWR	voltage standing wave ratio	SWR	standing wave ratio
VNA	vector network analyzer	SA	spectrum analyzer
DUT	device under test	S-params	scattering parameters
RL	return loss	IL	insertion loss
ML	mismatch loss	NF	noise figure
BW	bandwidth	SNR	signal to noise ratio
Tx, TX	transmitter	Rx, RX	receiver
BTS	base transceiver station	GSM	global system for mobile communications
EIRP	effective isotropic radiated power	ERP	effective radiated power
SAR	specific absorption rate	SARPs	standards and recommended practices
HERP	hazards of EM radiation to personnel	RADAR	radio detection and ranging
ASR	airport surveillance radar	CW	continuous wave

Short	Means	Short	Means
TWT, TWTA	traveling wave tube (amplifier)	BWO	backward wave oscillator
CFA	cross-field amplifier	LNA	low noise amplifier
HPA	high power amplifier	SWS	slow wave structure
O-type	linear beam tube (axial B -field)	M-type	crossed-field tube ($E \perp B$)
MASER	microwave amplification by stimulated emission of radiation	LASER	light amplification by stimulated emission of radiation
MESFET	metal-semiconductor FET	HEMT	high electron mobility transistor
GaAs	gallium arsenide	FET	field effect transistor
BJT	bipolar junction transistor	PIN	p-type, intrinsic, n-type diode
IMPATT	impact ionization avalanche transit-time	TRAPATT	trapped plasma avalanche triggered transit
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
MAG	maximum available gain	MSG	maximum stable gain
MIC	microwave integrated circuit	MMIC	monolithic microwave integrated circuit
ADS	advanced design system	IF	intermediate frequency
LO	local oscillator	RFIC	radio frequency integrated circuit
PCB	printed circuit board	SMA	subminiature version A connector

Short	Means	Short	Means
attr(s)	attribute(s)	no.	number
freq	frequency	info	information
ops	operations	config	configuration
rm	remove	vs, Vs	versus
w/, w/o	with, without	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and
=	is / equals	norm	normalized
max, min	maximum, minimum	calc	calculation / calculate
eq, eqn	equation	approx	approximately
pts	points	coeff	coefficient
temp	temperature	params	parameters

Standard and component terms (proper names and symbols, kept as standard): **Magic Tee** 180° hybrid junction (E-plane & H-plane tee combined) · **Directional Coupler** 4-port forward/reverse coupled wave device · **Circulator** non-reciprocal multi-port ferrite device ($1 \rightarrow 2 \rightarrow 3 \rightarrow 1$) · **Gunn Diode** transferred electron device (GaAs / InP, negative resistance) · **Bolometer** power detector (barretter: PTC, thermistor: NTC) · **Calorimeter** power measurement via heat transfer (static, circulating, dry, flow).

Key mathematical symbols: Z_0 characteristic impedance · Y_0 characteristic admittance · Γ reflection coefficient · S scattering matrix · K Rollett's stability factor · Δ determinant of S-matrix · μ single parameter stability metric · G_T transducer gain · G_{TU} unilateral transducer gain · f_c cutoff frequency · λ_g guide wavelength · v_p phase velocity · v_g group velocity.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams.

Monospace year tags are BEI (EX 716, IV/I); normal font tags are BEX (EX 752 / EG 785EX, IV/II).

Tier chips over 24 papers, 2069–2082 BS: **TOP /24** = topic asked in 7 or more papers · **HOT /24** = 3 to 6 papers · **PIN /24** = 1 or 2 papers but worth ≥ 6 marks.

Mark chips such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.